```sql
set enable_seqscan = off;
explain (analyze, buffers)
with category_stats as (
   select
       p.product_category,
       count(*) as orders_count
   from opt_orders o
   join opt_clients c
       on c.id = o.client_id
   join opt_products p
       on p.product_id = o.product_id
   where c.status = 'active'
     and o.order_date >= date '2023-01-01'
     and o.order_date < date '2024-01-01'
   group by p.product_category
)
select
   (
       select product_category
       from category_stats
       order by orders_count desc, product_category asc
       limit 1
   ) as most_popular_category,
   (
       select product_category
       from category_stats
       order by orders_count asc, product_category asc
       limit 1
   ) as least_popular_category,
   (
       select avg(orders_count)
       from category_stats
   ) as avg_orders_per_category;
reset enable_seqscan
```

QUERY PLAN
Result (cost=10277.28..10277.29 rows=1 width=268) (actual time=455.201..471.890 rows=1.00 loops=1)
Buffers: shared hit=1357

CTE category_stats
-> Finalize GroupAggregate (cost=10275.39..10276.90 rows=5 width=18) (actual time=452.452..469.154 rows=5.00 loops=1)
Group Key: p.product_category
Buffers: shared hit=1357
-> Gather Merge (cost=10275.39..10276.79 rows=12 width=18) (actual time=452.431..469.130 rows=15.00 loops=1)
Workers Planned: 2
Workers Launched: 2
Buffers: shared hit=1357
-> Sort (cost=9275.37..9275.38 rows=5 width=18) (actual time=211.060..211.068 rows=5.00 loops=3)
Sort Key: p.product_category
Sort Method: quicksort Memory: 25kB
Buffers: shared hit=1357
Worker 0: Sort Method: quicksort Memory: 25kB
Worker 1: Sort Method: quicksort Memory: 25kB
-> Partial HashAggregate (cost=9275.26..9275.31 rows=5 width=18) (actual time=209.897..209.913 rows=5.00 loops=3)
Group Key: p.product_category
Batches: 1 Memory Usage: 32kB
Buffers: shared hit=1341
Worker 0: Batches: 1 Memory Usage: 32kB
Worker 1: Batches: 1 Memory Usage: 32kB
-> Hash Join (cost=2033.92..9069.64 rows=41124 width=10) (actual time=38.588..193.574 rows=33289.00 loops=3)
Hash Cond: (o.product_id = p.product_id)
Buffers: shared hit=1341

-> Parallel Hash Join (cost=1956.87..8884.19 rows=41124 width=4) (actual time=33.620..173.106 rows=33289.00 loops=3)
Hash Cond: (o.client_id = c.id)
Buffers: shared hit=1240
-> Parallel Index Only Scan using idx_opt_orders_date_client_product on opt_orders o (cost=0.42..6711.90 rows=82220 width=20) (actual time=0.417..90.850 rows=66568.33 loops=3)
Index Cond: ((order_date >= '2023-01-01'::date) AND (order_date < '2024-01-01'::date))
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=990
-> Parallel Hash (cost=1695.95..1695.95 rows=20840 width=16) (actual time=32.637..32.637 rows=16674.67 loops=3)
Buckets: 65536 Batches: 1 Memory Usage: 2880kB
Buffers: shared hit=250
-> Parallel Index Only Scan using idx_opt_clients_status_id on opt_clients c (cost=0.42..1695.95 rows=20840 width=16) (actual time=1.747..59.040 rows=50024.00 loops=1)
Index Cond: (status = 'active'::text)
Heap Fetches: 0
Index Searches: 1
Buffers: shared hit=250
-> Hash (cost=64.54..64.54 rows=1000 width=14) (actual time=4.757..4.758 rows=1000.00 loops=3)
Buckets: 1024 Batches: 1 Memory Usage: 55kB
Buffers: shared hit=101
-> Index Scan using opt_products_pkey on opt_products p (cost=0.28..64.54 rows=1000 width=14) (actual time=0.505..4.128 rows=1000.00 loops=3)

Index Searches: 3
Buffers: shared hit=101
InitPlan 2
-> Limit (cost=0.13..0.13 rows=1 width=126) (actual time=453.158..453.161 rows=1.00 loops=1)
Buffers: shared hit=945
-> Sort (cost=0.13..0.14 rows=5 width=126) (actual time=453.156..453.157 rows=1.00 loops=1)
Sort Key: category_stats.orders_count DESC, category_stats.product_category
Sort Method: top-N heapsort Memory: 25kB
Buffers: shared hit=945
-> CTE Scan on category_stats (cost=0.00..0.10 rows=5 width=126) (actual time=452.461..452.486 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
Buffers: shared hit=945
InitPlan 3
-> Limit (cost=0.13..0.13 rows=1 width=126) (actual time=0.262..0.264 rows=1.00 loops=1)
-> Sort (cost=0.13..0.14 rows=5 width=126) (actual time=0.158..0.159 rows=1.00 loops=1)
Sort Key: category_stats_1.orders_count, category_stats_1.product_category
Sort Method: top-N heapsort Memory: 25kB
-> CTE Scan on category_stats category_stats_1 (cost=0.00..0.10 rows=5 width=126) (actual time=0.011..0.013 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
InitPlan 4
-> Aggregate (cost=0.12..0.12 rows=1 width=32) (actual time=0.925..0.937 rows=1.00 loops=1)

-> CTE Scan on category_stats category_stats_2 (cost=0.00..0.10 rows=5 width=8) (actual time=0.001..0.002 rows=5.00 loops=1)
Storage: Memory Maximum Storage: 17kB
Planning:
Buffers: shared hit=11
Planning Time: 31.362 ms
Execution Time: 478.351 ms